

CS2601 Linear and Convex Optimization

Homework 13 Solution

Due: 2022.1.3

1. Consider the following optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}} \quad & f(x) = \log(1 + e^x) \\ \text{s. t.} \quad & x \geq 0 \end{aligned}$$

- (a). Find the optimal solution and the optimal value.
- (b). Find the dual function and the dual problem.
- (c). Find the dual optimal solution and the dual optimal value. Does strong duality hold?

Solution. (a). f is monotonically increasing, so $f^* = f(0) = \log 2$ and $x^* = 0$.

(b). The Lagrangian is

$$\mathcal{L}(x, \mu) = \log(1 + e^x) - \mu x$$

The dual function is

$$\phi(\mu) = \inf_x \mathcal{L}(x, \mu) = \inf_x [\log(1 + e^x) - \mu x]$$

Note

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{e^x}{1 + e^x} - \mu \in (-\mu, 1 - \mu)$$

If $\mu \in (0, 1)$, then

$$\frac{\partial \mathcal{L}}{\partial x} = 0 \implies x = \log \frac{\mu}{1 - \mu} \implies \phi(\mu) = \mathcal{L}(\log \frac{\mu}{1 - \mu}, \mu) = -\mu \log \mu - (1 - \mu) \log(1 - \mu)$$

If $\mu \leq 0$, then

$$\frac{\partial \mathcal{L}}{\partial x} > 0 \implies \phi(\mu) = \lim_{x \rightarrow -\infty} \mathcal{L}(x, \mu) = \begin{cases} 0, & \mu = 0 \\ -\infty, & \mu < 0 \end{cases}$$

If $\mu \geq 1$, then

$$\frac{\partial \mathcal{L}}{\partial x} < 0 \implies \phi(\mu) = \lim_{x \rightarrow +\infty} \mathcal{L}(x, \mu) = -\infty$$

Therefore, the dual function is

$$\phi(\mu) = \begin{cases} \mathcal{L}(\log \frac{\mu}{1-\mu}, \mu) = -\mu \log \mu - (1-\mu) \log(1-\mu), & 0 < \mu < 1 \\ 0, & \mu = 0 \text{ or } \mu = 1 \\ -\infty, & \mu > 1 \text{ or } \mu < 0 \end{cases}$$

or, using the convention $0 \log 0 = 0$,

$$\phi(\mu) = \begin{cases} \mathcal{L}(\log \frac{\mu}{1-\mu}, \mu) = -\mu \log \mu - (1-\mu) \log(1-\mu), & 0 \leq \mu \leq 1 \\ -\infty, & \mu > 1 \text{ or } \mu < 0 \end{cases}$$

The dual problem is

$$\begin{aligned} \max \quad & \phi(\mu) = \begin{cases} \mathcal{L}(\log \frac{\mu}{1-\mu}, \mu) = -\mu \log \mu - (1-\mu) \log(1-\mu), & 0 \leq \mu \leq 1 \\ -\infty, & \mu > 1 \text{ or } \mu < 0 \end{cases} \\ \text{s. t.} \quad & \mu \geq 0 \end{aligned}$$

(c). For $\mu \in (0, 1)$,

$$\phi'(\mu) = -\log \mu + \log(1-\mu) = 0 \implies \mu = \frac{1}{2}$$

so

$$\phi^* = \max\{\phi(0), \phi(\frac{1}{2})\} = \log 2$$

and $\mu^* = \frac{1}{2}$. Since $f^* = \phi^*$, strong duality holds.

Alternatively, you can use Slater's condition to show that strong duality holds, so $\phi^* = f^* = \log 2$. The dual optimal solution is the Lagrange multiplier corresponding to $x^* = 0$, so by the stationarity condition,

$$\mu^* = \frac{e^{x^*}}{1 + e^{x^*}} = \frac{1}{2}.$$

□

2. Consider the optimization problem in Problem 2 of Homework 10,

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^2} \quad & f(\mathbf{x}) = x_1^2 + x_2^2 \\ \text{s. t.} \quad & (x_1 - 1)^2 + (x_2 - 1)^2 \leq 1 \\ & (x_1 - 1)^2 + (x_2 + 1)^2 \leq 1 \end{aligned}$$

- Find the Lagrange dual function and the dual problem.
- Find the dual optimal value ϕ^* . Does strong duality hold?
- Does Slater's condition hold? What can you conclude about the necessity of Slater's condition for strong duality?

- (d). Is the dual optimal value ϕ^* attained by any dual feasible point? Is this expected, given the answer to Problem 2(b) of Homework 10?

Solution. (a). The Lagrangian is

$$\begin{aligned}\mathcal{L}(\mathbf{x}, \boldsymbol{\mu}) &= x_1^2 + x_2^2 + \mu_1[(x_1 - 1)^2 + (x_2 - 1)^2 - 1] + \mu_2[(x_1 - 1)^2 + (x_2 + 1)^2 - 1] \\ &= (1 + \mu_1 + \mu_2)x_1^2 - 2(\mu_1 + \mu_2)x_1 + (1 + \mu_1 + \mu_2)x_2^2 - 2(\mu_1 - \mu_2)x_2 + \mu_1 + \mu_2\end{aligned}$$

If $\mu_1 + \mu_2 \leq -1$, the paraboloid opens down, so

$$\phi(\boldsymbol{\mu}) = \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\mu}) = -\infty$$

If $\mu_1 + \mu_2 > -1$, the paraboloid opens up, and $\mathcal{L}(\mathbf{x}, \boldsymbol{\mu})$ is minimized by

$$\begin{cases} x_1 = \frac{\mu_1 + \mu_2}{1 + \mu_1 + \mu_2} \\ x_2 = \frac{\mu_1 - \mu_2}{1 + \mu_1 + \mu_2} \end{cases}$$

so

$$\phi(\boldsymbol{\mu}) = \frac{\mu_1 + \mu_2 - (\mu_1 - \mu_2)^2}{1 + \mu_1 + \mu_2}$$

Thus the dual function is

$$\phi(\boldsymbol{\mu}) = \begin{cases} \frac{\mu_1 + \mu_2 - (\mu_1 - \mu_2)^2}{1 + \mu_1 + \mu_2}, & \text{if } \mu_1 + \mu_2 > -1 \\ -\infty, & \text{if } \mu_1 + \mu_2 \leq -1 \end{cases}$$

The dual problem is

$$\begin{aligned} \max_{\boldsymbol{\mu} \in \mathbb{R}^2} \quad & \phi(\boldsymbol{\mu}) = \begin{cases} \frac{\mu_1 + \mu_2 - (\mu_1 - \mu_2)^2}{1 + \mu_1 + \mu_2}, & \text{if } \mu_1 + \mu_2 > -1 \\ -\infty, & \text{if } \mu_1 + \mu_2 \leq -1 \end{cases} \\ \text{s.t.} \quad & \boldsymbol{\mu} \geq \mathbf{0} \end{aligned}$$

which is equivalent to

$$\begin{aligned} \max_{\boldsymbol{\mu} \in \mathbb{R}^2} \quad & \frac{\mu_1 + \mu_2 - (\mu_1 - \mu_2)^2}{1 + \mu_1 + \mu_2} \\ \text{s.t.} \quad & \boldsymbol{\mu} \geq \mathbf{0} \end{aligned}$$

- (b). By weak duality,

$$\phi^* \leq f^* = 1.$$

When $\mu_1 = \mu_2 = \mu \rightarrow \infty$,

$$\phi(\mu, \mu) = \frac{2\mu}{1 + 2\mu} \rightarrow 1,$$

so

$$\phi^* = \sup_{\boldsymbol{\mu} \geq \mathbf{0}} \phi(\boldsymbol{\mu}) = 1 = f^*$$

Strong duality holds.

(c). Slater's condition does not hold, as both constraints hold with equality at the unique feasible point $\mathbf{x}^*$. This shows that Slater's condition is not necessary for strong duality to hold.

(d). Note

$$\phi(\boldsymbol{\mu}) = \frac{\mu_1 + \mu_2 - (\mu_1 - \mu_2)^2}{1 + \mu_1 + \mu_2} \leq \frac{\mu_1 + \mu_2}{1 + \mu_1 + \mu_2} < 1$$

for any $\boldsymbol{\mu} \geq \mathbf{0}$, so the dual optimal value ϕ^* is not attained by any dual feasible point.

This is expected. Note the primal problem is convex, there exists a primal optimal solution $\mathbf{x}^*$, and strong duality holds. By Problem 2(b) of Homework 10, there are no Lagrange multipliers satisfying the KKT conditions at $\mathbf{x}^*$. Thus the dual optimal solution cannot exist; otherwise, the KKT conditions would hold.

□

3. Consider the following minimization problem

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^2} \quad f(\mathbf{x}) &= \begin{cases} x_1^3 + x_2^3, & \text{if } \mathbf{x} \geq \mathbf{0} \\ +\infty, & \text{otherwise} \end{cases} \\ \text{s.t.} \quad x_1 + x_2 &\geq 1 \end{aligned} \tag{P1}$$

Note the domain of f is $\text{dom } f = \{\mathbf{x} \in \mathbb{R}^2 : \mathbf{x} \geq \mathbf{0}\}$ and the domain of the problem is $D = \text{dom } f$.

(a). Since D is not the entire space, the dual function of this problem is defined by

$$\phi(\mu) = \inf_{\mathbf{x} \in D} \{f(\mathbf{x}) + \mu(1 - x_1 - x_2)\}$$

Find the explicit expression of $\phi(\mu)$.

(b). Find the dual optimal solution.

(c). What is the primal optimal value? Hint: Note f is convex on its domain.

(d). Note the primal problem (P1) is equivalent to

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^2} \quad f_1(\mathbf{x}) &= x_1^3 + x_2^3 \\ \text{s.t.} \quad x_1 + x_2 &\geq 1 \\ \mathbf{x} &\geq \mathbf{0} \end{aligned} \tag{P2}$$

What's the dual function of this equivalent problem (P2)? Does strong duality hold for (P2)?

Remark. Note $\text{dom } f_1 = \mathbb{R}^2$ and f_1 is not convex. This problem shows that equivalent primal problems can have very different dual problems. Not all dual problems are equally useful.

Solution. (a). Since D is not the entire space, the dual function of this problem is defined by

$$\begin{aligned}\phi(\mu) &= \inf_{x_1 \geq 0, x_2 \geq 0} \{x_1^3 + x_2^3 + \mu(1 - x_1 - x_2)\} \\ &= \inf_{x_1 \geq 0} \{x_1^3 - \mu x_1\} + \inf_{x_2 \geq 0} \{x_2^3 - \mu x_2\} + \mu \\ &= \begin{cases} -4(\frac{\mu}{3})^{3/2} + \mu, & \text{if } \mu \geq 0 \\ \mu, & \text{if } \mu \leq 0 \end{cases}\end{aligned}$$

(b). For $\mu > 0$,

$$\phi''(\mu) = -\frac{1}{3}\sqrt{\frac{3}{\mu}} < 0$$

so ϕ is concave on $(0, +\infty)$.

$$\phi'(\mu) = 1 - 2\sqrt{\frac{\mu}{3}} = 0 \implies \mu = \frac{3}{4}$$

So

$$\max_{\mu > 0} \phi(\mu) = \phi(\frac{3}{4}) = \frac{1}{4}$$

and

$$\phi^* = \max_{\mu \geq 0} \phi(\mu) = \max \left\{ \phi(0), \phi(\frac{3}{4}) \right\} = \frac{1}{4}$$

(c). Slater's condition holds. For example, $\mathbf{x} = (1, 1) \in \text{int}D$ and $1 - x_1 - x_2 < 0$. So $f^* = \phi^* = \frac{1}{4}$.

(d). The dual function is

$$\phi_1(\boldsymbol{\mu}) = \inf_{\mathbf{x}} \{x_1^3 + x_2^3 - \mu_1(x_1 + x_2 - 1) - \mu_2 x_1 - \mu_3 x_3\} = -\infty$$

So the dual optimal value is $\phi_1^* = -\infty \neq f_1^* = f^* = \frac{1}{4}$. Strong duality does not hold for (P2).

□

4. Hard-margin SVM. Recall the primal formulation of the hard-margin SVM is

$$\begin{aligned} \min_{\mathbf{w}, b} \quad & \frac{1}{2} \|\mathbf{w}\|_2^2 \\ \text{s. t.} \quad & y_i(\mathbf{x}_i^T \mathbf{w} + b) \geq 1, \quad i = 1, 2, \dots, n \end{aligned}$$

and the dual formulation is

$$\begin{aligned} \max_{\boldsymbol{\mu}} \quad & \mathbf{1}^T \boldsymbol{\mu} - \frac{1}{2} \boldsymbol{\mu}^T \mathbf{Q} \boldsymbol{\mu} \\ \text{s. t.} \quad & \boldsymbol{\mu}^T \mathbf{y} = 0 \\ & \boldsymbol{\mu} \geq \mathbf{0} \end{aligned}$$

where

$$Q_{ij} = y_i y_j \mathbf{x}_i^T \mathbf{x}_j = (y_i \mathbf{x}_i)^T (y_j \mathbf{x}_j).$$

The primal optimal $\mathbf{w}^*$ can be obtained from the dual optimal solution μ^* by

$$\mathbf{w}^* = \sum_{i=1}^n \mu_i^* y_i \mathbf{x}_i$$

(a). Show that for any i with $\mu_i^* > 0$,

$$y_i(\mathbf{x}_i^T \mathbf{w}^* + b^*) = 1,$$

and hence

$$b^* = y_i - \mathbf{x}_i^T \mathbf{w}^*$$

(b). In this problem, we use projected gradient descent to solve the dual problem and then recover the primal optimal solutions $\mathbf{w}^*$ and b^* . Complete the implementation in `svm.py`. For the projection `proj`, use your implementation in Problem 4(b) of Homework 10. Show the output.

Solution. (a). By the complementary slackness condition,

$$\mu_i[y_i(\mathbf{x}_i^T \mathbf{w}^* + b^*) - 1] = 0$$

If $\mu_i > 0$, then

$$y_i(\mathbf{x}_i^T \mathbf{w}^* + b^*) = 1,$$

Since $y_i \in \{+1, -1\}$,

$$\mathbf{x}_i^T \mathbf{w}^* + b^* = y_i$$

and hence

$$b^* = y_i - \mathbf{x}_i^T \mathbf{w}^*$$

(b). `primal optimal:`

```
w = [-1.09090908  1.45454545]
b = -0.09090909915526812
```

`dual optimal:`

```
mu = [1.65289255e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00
0.00000000e+00 0.00000000e+00 0.00000000e+00 1.65289254e+00
0.00000000e+00 0.00000000e+00 7.11813918e-09 0.00000000e+00
0.00000000e+00]
```

